

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

AU-6002-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Automotive Design and Assembly Drawing

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions. All questions carry equal marks.
ii) All parts of each question are to be attempted at one place.
iii) Draw neat and clean sketches/diagrams/figures wherever required.
1. a) Explain the stages involved in the design of a machine element. 7
b) List the factors which affect the friction between two contacting surfaces. 7
 2. a) Explain the design of Double Dwell Cam. 7
b) Explain the design of Single Shoe Brake. 7
 3. a) Explain Energy absorption and heat dissipation through brake. 7
b) Define Pressure Angle and Radius of curvature in cam motion. 7
 4. a) Explain and design of suspension spring and also define Front axle. 7

AU-6002-CBGS

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[2]

- b) A closed coil helical spring is having 15 coils and average diameter of spring is 70mm. The diameter of spring wire is 12mm. Calculate applied load when deflection is 22mm and applied load when maximum shear stress is 170 MPa, assume G is $81 \times 10^3 \text{ MN/m}^2$. 7
5. a) List the principal parts of an IC engine and also explain briefly. 7
- b) A railway axle made up of mild steel and a 80kN load is acting on it. The wheels are at 1.6m apart and the bogie is fixed on the axle on a spring at a 1.1m distance. Assume the permissible stress of the axle material is 70 MN/m^2 , then calculate the diameter of the axle. 7
6. Design a side crankshaft for a $500 \text{ mm} \times 600 \text{ mm}$ gas engine. The weight of the fly wheel is 80kN and the explosion pressure is 2.5 N/mm^2 . The gas pressure at maximum torque is 0.9 N/mm^2 , when crank angle is 30° . The connecting rod is 4.5 times the crank radius. Any other data required for design may be assumed. 14
7. Design a leaf spring for the following specifications: 14
Total load = 140 kN, Maximum number of leaves = 10
Span of spring = 1000mm, Permissible deflection = 80mm
Take Young's modulus $E = 2 \times 10^5 \text{ N/mm}^2$ and allowable stress in spring material as 600 N/mm^2 .
8. Write a notes on Any Four of the following: 14
- a) SOHC and DOHC
 - b) Connecting rod assembly
 - c) Disc Brake
 - d) Shackle assembly
 - e) Theories of Failure

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